

Repeating Questions of last Final Exam (Held in winter 2026)

Course: Zoo 512T

- **Insects: Metamorphosis & Hormones**

Q: Define metamorphosis.

Metamorphosis is the process of developmental change in which an insect transforms from an immature stage into an adult through distinct stages.

Q: What is holometabolous metamorphosis?

It is complete metamorphosis with four stages: egg → larva → pupa → adult.

Q: What is paurometabolous metamorphosis?

It is gradual metamorphosis in which the young nymph resembles the adult but is smaller and lacks functional wings and mature reproductive organs.

Q: What are the two main hormones in insects?

The two main hormones are juvenile hormone (JH) and ecdysone.

Q: Role of juvenile hormone in metamorphosis.

Juvenile hormone maintains juvenile characteristics during moulting. High JH levels cause immature stages to remain juvenile, while a decrease in JH allows metamorphosis toward the adult stage.

Q: How many parts is the insect body divided into?

The insect body is divided into three main parts: head, thorax, and abdomen.

Q: Name the body parts of an insect.

The three main body regions are head, thorax, and abdomen. The thorax bears three pairs of legs and usually one or two pairs of wings.

Q: Explain mating behavior in insects.

Insect mating usually involves courtship, recognition between male and female, mating, and transfer of sperm. Courtship may involve visual signals, sounds, pheromones, or specific movements.

Q: What is insect pollination?

Insect pollination is the transfer of pollen grains from the anther of one flower to the stigma of the same or another flower by insects such as bees and butterflies.

Q: What is asynchronous flight?

Asynchronous flight occurs when insect wing muscles contract at a frequency different from the nerve impulses. A single nerve impulse can produce several muscle contractions, allowing very rapid wing beats.

Q: What is the ventilating mechanism in insects?

Insects ventilate their tracheal system by rhythmic movements of the abdomen and body. These movements pump air through spiracles into the tracheae and tracheoles, supplying oxygen directly to tissues and removing carbon dioxide.

Q: Importance of soil-dwelling insects.

Soil-dwelling insects help in decomposition, nutrient cycling, soil aeration, and formation of soil structure. Some also act as predators or pests.

Q: What are insect castes?

In social insects, castes are specialized groups with different functions, such as queen, males, and workers. In some species, soldiers are also present.

- **2. Crayfish**

Q: What are the sensory structures of crayfish?

The main sensory structures are antennae, antennules, compound eyes, statocysts, and sensory setae.

Q: Describe the circulatory system of crayfish.

Crayfish has an open circulatory system. A dorsal heart pumps hemolymph into arteries, which release it into body spaces called sinuses. The hemolymph then returns to the heart through openings called ostia.

Q: Describe the nervous system of crayfish.

The nervous system consists of a brain (supraesophageal ganglion), circumoesophageal connectives, subesophageal ganglion, and a double ventral nerve cord with segmental ganglia.

Q: How does gaseous exchange occur in crayfish?

Gaseous exchange occurs through gills. Water passes over the gills, where oxygen diffuses into the hemolymph and carbon dioxide diffuses out.

- **3. Echinoderms**

Q: What are the characteristics of phylum Echinodermata?

Echinoderms are exclusively marine, have spiny skin, calcareous endoskeleton, water vascular system, tube feet, and pentaradial symmetry in adults. They are triploblastic, coelomate, and deuterostomes.

Q: What are ossicles?

Ossicles are small calcareous skeletal elements embedded in the dermis of echinoderms, providing support and protection.

Q: What is the function of tube feet in echinoderms?

Tube feet are used for locomotion, attachment, feeding, respiration, and sensory functions.

Q: What are dermal branchiae?

Dermal branchiae, or skin gills, are thin-walled projections of the body wall that help in respiration and removal of metabolic wastes.

Q: What is Aristotle's lantern?

Aristotle's lantern is a complex jaw-like feeding apparatus of sea urchins, used mainly for scraping and grazing food from surfaces.

Q: What is the role of papillae in sea urchins?

Papillae are small projections of the body wall that assist in respiration and removal of waste materials.

Q: How does regeneration occur in starfish?

Starfish regenerate lost body parts through cell division and tissue differentiation. A lost arm can regenerate, and in some species a detached arm with part of the central disc can develop into a new individual.

Q: Difference between brittle star and basket star.

Brittle Star

Basket Star

Arms are usually slender and simple.

Arms are highly branched.

Moves mainly by arm movements.

Uses branched arms for feeding and collecting food.

Usually has five simple arms.

Arms repeatedly branch into many branches.

Q: What is the stomach of a starfish?

The starfish stomach has two parts: the cardiac stomach and the pyloric stomach. The cardiac stomach can be everted to digest food outside the body.

Q: What is trepang?

Trepang is the dried body of a sea cucumber, commonly used as food and known as sea cucumber or beche-de-mer.

Q: Describe locomotion and movement of sea cucumber.

Sea cucumbers move using tube feet and muscular contractions of the body wall. Some species also use their oral tube feet for movement.

Q: Describe development and reproduction of echinoids.

Echinoids generally have separate sexes. Fertilization is external, and development includes a free-swimming echinopluteus larva, which later undergoes metamorphosis into the adult.

- **4. Sponges**

Q: What are the body forms of sponges?

The three main body forms of sponges are asconoid, syconoid, and leuconoid. They differ mainly in the complexity of their canal systems.

- **5. Annelids**

Q: What are the characteristics of annelids?

Annelids are bilaterally symmetrical, triploblastic, coelomate, segmented worms. They have a complete digestive tract, closed circulatory system, and organ-system level of organization.

- **6. Platyhelminthes & Nematodes**

Q: What is the body wall of turbellarians?

The body wall consists of an outer epidermis, basement membrane, and layers of circular, longitudinal, and diagonal muscles. The epidermis bears cilia in many turbellarians.

Q: What are the evolutionary adaptations of nematodes?

Nematodes have a tough cuticle, pseudocoel, complete digestive tract, specialized feeding structures, and high reproductive capacity, which help them survive in diverse habitats, including parasitic environments.

Q: What diseases are caused by Ascaris?

Ascaris lumbricoides causes ascariasis, which may cause abdominal discomfort, intestinal blockage, and nutritional problems.

- **7. Other Important Short Questions**

Q: What are Malpighian tubules?

Malpighian tubules are excretory organs of insects that remove nitrogenous wastes, mainly uric acid, from the hemolymph and help regulate water and ion balance.

Q: What are book lungs?

Book lungs are respiratory organs found mainly in spiders and other arachnids. They consist of numerous thin air-filled plates arranged like the pages of a book.

Q: What is a garden centipede?

A garden centipede is a small, fast-moving centipede commonly found in soil and leaf litter. It is a predatory arthropod that feeds on small insects and other invertebrates.

Q: What is zonitis?

Zonitis is a genus of blister beetles belonging to the family Meloidae.

Q: What is a seta?

A seta is a bristle-like structure arising from the body wall, commonly found in annelids and used mainly for locomotion or anchorage.

Q: What is an annulus?

An annulus means a ring-like structure or circular groove.

Q: What is the phylogenetic consideration of Arthropoda?

Arthropods are considered closely related to other protostome invertebrates, especially annelids, based on similarities in segmentation and development, although their exact evolutionary relationship has been debated.